

Review Form of the ICCC 2026

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Paper ID	IC069	
Paper Title	Coverage-Oriented Planner-Guided Memory Residual Reinforcement Learning for Robotic Thermal-Spot Coverage	
Paper Score: Review Content and Score (full mark: 60)		
●	(Poor: 1-3; Fair: 4-6; Good: 7-8; Excellent: 9-10)	
Relevance	7 / 10	
Originality	6 / 10	
Significance	5 / 10	
Technical soundness	5 / 10	
Clarity of presentation/ language	6 / 10	
Match to Conference Topic	9 / 10	
Total Score:	38 / 60	
Comments and Instructions (better no less than 100 words):		

High-Level Assessment

This manuscript studies planner-guided residual reinforcement learning for dynamic hotspot coverage in a robotic thermal-processing setting. The topic is directly within ICCV 2026 Track 6 Robotics and also aligns with AI for Engineering and Machine Learning. The strongest methodological features are held-out evaluation seeds, a reported confidence interval for the proposed method versus Horizon-2, and explicit disclosure that the risk-aware heuristic can slightly exceed coverage while the proposed policy incurs higher latency and backlog in the Extreme regime. Those statements appropriately bound the claim; the paper presents a configurable coverage-oriented operating point rather than universal superiority. However, substantial reproducibility and comparison-validity gaps remain. None of the central numerical hyperparameters is reported; Horizon-2 and the risk-aware heuristic are not sufficiently defined; the masked-attention, LSTM, and PPO architecture and training budget are incomplete; and the dramatic collapse of vanilla LSTM-PPO raises a serious baseline-fairness question. The resampling unit for the nested training-seed, evaluation-seed, and episode design is unclear, Table II lacks uncertainty for the heuristic and ablation baselines, and contribution attribution among vanilla, no-residual, and residual variants is internally difficult to reconcile. These issues are substantial but appear repairable through complete specification, hierarchical uncertainty reporting, fair baseline tuning, a variant matrix, and a compact planner-mismatch test. I therefore rate the paper Fair and recommend Review Again After Required Major Changes.

1. Does the introduction state the purpose of the paper?

Yes. The Introduction appears to state a deliberately narrow purpose: use a receding-horizon planner as a structured action prior, let a residual PPO policy adapt that plan to dynamic hotspot arrivals, and evaluate the resulting coverage-service trade-off rather than claim a universal replacement for planning. That purpose is relevant and understandable. However, the manuscript should explicitly identify its ICCV positioning under Track 6 Robotics, with secondary alignment to Track 2 AI for Engineering and Track 4 Machine Learning. It should also instantiate the physical setting more concretely. 'Dynamic thermal processing' is too abstract unless the authors identify the intended process class, robot or motion platform, sensing source for hotspot locations, service action, arrival model, and operational meaning of coverage, latency, SLA, backlog, and smoothness. The Introduction should end with two or three testable questions: whether residual learning improves a fixed planner prior, whether the advantage persists under a different planner horizon, and how coverage gains trade against service delay and backlog.

2. Are the references relevant and complete? Supply missing references (please specify).

The current references are relevant to reinforcement learning, residual policies, and planning-guided control, but the literature positioning is incomplete in three areas. First, the post-hoc action filter is most directly related to M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, and U. Topcu, 'Safe Reinforcement Learning via Shielding,' Proceedings of AAAI, vol. 32, no. 1, 2018. Second, if the authors wish to formulate latency or backlog as explicit constraints rather than reward weights, they should discuss J. Achiam, D. Held, A. Tamar, and P. Abbeel, 'Constrained Policy Optimization,' Proceedings of ICML, PMLR 70, pp. 22-31, 2017, while clearly distinguishing CMDP optimization from the present shielded, scalarized-reward implementation. Third, the task is closely related to the dynamic traveling repairman problem and stochastic online service routing. The authors should add D. J. Bertsimas and G. van Ryzin, 'A Stochastic and Dynamic Vehicle Routing Problem in the Euclidean Plane,' Operations Research, vol. 39, no. 4, pp. 601-615, 1991, and their multi-vehicle extension in Operations Research, vol. 41, no. 1, pp. 60-76, 1993. These references provide the strongest classical context and potential nonlearning baselines for stochastic arrivals and waiting-time objectives.

3. Is the paper clearly written and well organized?

The manuscript is generally organized around the architecture, aggregate comparison, ablations, representative trajectory, and reward sweep, and it commendably states where the risk-aware heuristic performs better and where the proposed method pays a latency or backlog cost. However, clarity is reduced by extensive repetition of the Extreme-regime interpretation, while load-bearing implementation details remain absent. The six reward coefficients, four risk-score weights and feature functions, β_t schedule, k_{phase} , guard thresholds, r_{cover} , T_{SLA} , A_{norm} , network dimensions, and PPO hyperparameters should be reported. Horizon-2 and the risk-aware heuristic require complete algorithmic definitions rather than names alone. A compact variant table should state exactly how vanilla LSTM-PPO, no-residual PPO, Horizon-2, the heuristic, and the full method differ. The conclusion's paragraph describing which figures and tables were retained for page layout should be deleted because it is editorial meta-commentary rather than scientific content. Fig. 2 should also be harmonized with Table I: if the rounded difference is -0.000, either explain the rounding or use a consistent 'approximately tied' label.

4. Does the author explain the significance of this paper?

Yes, at the level of a simulation-based architecture and evaluation study. The paper addresses a meaningful robotics problem with stochastic arrivals and service congestion, and it provides a useful result that residual learning can improve the Horizon-2 operating point in Extreme conditions while incurring higher latency and backlog. Its explicit statement that the risk-aware heuristic slightly exceeds coverage in some regimes is appropriate claim calibration, not a weakness by itself. The paper should not be required to prove superiority over that heuristic when it has not made such a claim. Nevertheless, the significance score should remain moderate because the strongest nonlearning alternative is undefined, its apparent advantage has no confidence interval, and the practical reason to prefer a much more complex learned pipeline is not yet demonstrated. A real-robot experiment or full sim-to-real study would strengthen the work but should be treated as a recommended extension rather than a mandatory conference threshold. A lower-cost, high-value experiment is to replace Horizon-2 with Horizon-1 or Horizon-3 and show whether the residual-policy conclusion is planner-independent.

5. What is the contribution of the paper? Are the research contributions clear?

The high-level contributions are identifiable: (i) a receding-horizon planner used as an action prior; (ii) a residual PPO policy with a masked set encoder and temporal state; (iii) action clipping and direction/progress guards; (iv) a reward-based coverage, latency, backlog, SLA, and smoothness trade-off; and (v) evaluation over multiple load stages with held-out seeds and a reported interval for the Extreme comparison against Horizon-2. The paper also correctly limits its claim to a coverage-oriented operating point rather than universal dominance. However, the contribution attribution is not yet reliable. In Extreme, vanilla LSTM-PPO reportedly scores 0.061, no-residual PPO 0.687, and the full method 0.855. The first gap is much larger than the residual-interface gap, yet the text reportedly says that removing attention or other auxiliary components changes some stages by only a few thousandths. The authors must provide an exact component matrix for all variants and revise the causal narrative. Otherwise, it is unclear whether the main gain comes from observation encoding, reward design, training choices, planner residualization, or several simultaneous changes.

6. If the paper is not technically sound, why not? How to improve it?

The method is plausible, and Technical Soundness in the Fair range is appropriate. The main problems are reproducibility, baseline fairness, statistical dependence, and incomplete attribution rather than a demonstrated impossibility of the approach.

(a) Essential - complete method specification. Report all six reward coefficients, all risk-score weights, the definitions and normalization of `phi_age`, `phi_dist`, `phi_mat`, and `phi_reach`, the `beta_t` warm-up schedule, `k_phase`, action bounds, direction/progress guard thresholds, `r_cover`, `T_SLA`, `A_norm`, masked-attention and LSTM dimensions, PPO hyperparameters, training steps, curriculum, normalization, and checkpoint-selection rule.

(b) Essential - define the comparators. Give pseudocode, objective, horizon, candidate-action generation, cost terms, and tie-breaking rules for Horizon-2. Fully define the risk-aware heuristic and verify its non-monotonic stage pattern: 0.924, 0.901, 0.836, and 0.863. Because this method is used as the strongest nonlearning reference, its implementation cannot remain implicit.

(c) Essential - baseline fairness. The very low vanilla LSTM-PPO coverage may reflect observation scaling, action parameterization, reward design, or an inadequate tuning budget rather than an inherent failure of direct RL. Report identical training budgets, hyperparameter-search ranges, learning curves, normalization, network capacity, and action interfaces for all learned baselines. Either retune the baseline under a disclosed protocol or substantially weaken conclusions based on its collapse.

(d) Essential - variant accounting and attribution. Provide a table showing the exact components present in vanilla, no-residual, each auxiliary ablation, and the full method. Reconcile the large vanilla-to-no-residual improvement with the claim that removing attention, carry, prediction, or service shaping changes performance only marginally. Do not assign the whole gain to residual planning unless the comparisons isolate that factor.

(e) Essential - uncertainty and dependence. The design contains three training seeds, three evaluation seeds, and five episodes per combination. State the statistical unit and use seed-level or hierarchical resampling that preserves nested dependence. Report results by training seed and add SD or confidence intervals for the risk-aware heuristic, no-residual PPO, and vanilla LSTM-PPO. Separate training variability from evaluation-environment variability when comparing the stochastic learned policy with deterministic Horizon-2.

(f) Essential - complete metric reporting. Fig. 1 and the reward formula refer to SLA and smoothness, but the manuscript does not report an SLA metric, smoothness values, or the numerical `T_SLA`. Report these outcomes or remove them from the evidentiary claims. Explain the coverage denominator, service completion rule, backlog definition, p90-latency construction, and aggregation across episodes and stages.

(g) Targeted validation and claim calibration. A full hardware study, broad sim-to-real analysis, and four-dimensional robustness sweep are desirable but not all necessary as an ICCC minimum. At least add a planner-mismatch test using another horizon or receding-horizon policy, and, if feasible, one unseen arrival-rate or layout condition. Preserve the paper's existing bounded claim: the method offers one configurable coverage-oriented architecture choice and is not uniformly faster or universally superior to the heuristic.

7. If the paper is too long for its category, how can it be shortened? Which part is not necessary?

The paper is not too long. It is reportedly five pages, which is the ICCC minimum, while the official submission rules permit up to ten pages with an additional-page fee beyond five. The authors therefore do not need to choose between deleting repetition and adding essential details; both can be done. Repeated explanations of why the Extreme stage matters should be condensed to one discussion paragraph, and the conclusion's page-layout meta-narrative should be removed. The recovered space and additional pages should be used for a complete hyperparameter table, Horizon-2 and heuristic pseudocode, a variant-component matrix, training-budget and learning-curve information, hierarchical uncertainty results, SLA and smoothness metrics, and a planner-mismatch experiment. Generic background on PPO or residual learning can be shortened, but implementation parameters, baseline definitions, and statistical units must remain in the main paper. The public ICCC submission guidance does not identify a supplementary-material route, so load-bearing reproducibility information should not be deferred to an assumed supplement.

Final Evaluation				
☐ Excellent	☐ Very Good	☐ Good	☐ Fair	☐ Poor
☐ Strongly Accept	☐ Accept	☐ Marginally Accept	☐ Review Again After Required Major Changes	☐ Reject